AllRatesToday

Currency Data API Specifications

Version 2.9 — October 2026

https://allratestoday.com

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1 Overview

The AllRatesToday Currency Data API is a RESTful web-services API that provides real-time and historical currency exchange rate data for over 160 world currencies. The API delivers mid-market rates aggregated from institutional foreign exchange market data, with no retail markup, plus the official published rates of 121 central banks and 3 tax authorities.

In addition to mid-market rates, the API publishes **official reference rates from 121 central banks** — including the US Federal Reserve, European Central Bank, Bank of England, People's Bank of China, and Bank of Japan — with full historical time series (11M+ data points), publication calendars, and CSV export. See Section 3, Central Bank Rates API.

This document provides a complete technical reference for integrating with the AllRatesToday API, including authentication, endpoints, parameters, response formats, error handling, and SDK usage across multiple programming languages.

1.1 Schema

- The API is accessed exclusively using the **HTTPS** protocol.
- The API is accessed from the domain `allratestoday.com`.
- All data is sent and received in **JSON** format by default.
- Keyless endpoints send **CORS headers** for browser use; keyed endpoints are called from your server (see 1.9).
- Responses are **Gzip compressed** for minimal bandwidth usage.

1.2 Base URL

All API requests should be made to the following base URL:

```
https://allratestoday.com/api/v1/
```

The API is deployed on Cloudflare's global edge network across 300+ locations, delivering sub-50ms response times worldwide.

1.3 Authentication

All requests to the authenticated API endpoints must include a valid API key. Authentication is performed via a **Bearer token** in the HTTP Authorization header.

Your API key is provided when you register for the service at:

```
https://allratestoday.com/register
```

API keys follow the format:

```
art_live_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

1.3.1 Authentication Example Using curl

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/rates?source=USD&amp;target=EUR"
```

1.3.2 Authentication Example Using JavaScript

```
const response = await fetch(
"https://allratestoday.com/api/v1/rates?source=USD&amp;target=EUR",
{
headers: {
"Authorization": "Bearer art_live_your_api_key"
}
}
);
```

1.3.3 Failed Authentication

Authenticating with invalid credentials will return **401 Unauthorized**:

```
HTTP/1.1 401 Unauthorized

{
"error": "Invalid or missing API key",
"docs": "https://allratestoday.com/docs"
}
```

1.4 Specifying the API Version

You specify the version of the API by including it in the URI path:

```
https://allratestoday.com/api/v{version_no}/...
```

Where `{version_no}` is an integer identifying the version. The current and only version is **v1**.

Example:

```
https://allratestoday.com/api/v1/rates?source=USD&amp;target=EUR
```

1.5 Data Format

The API returns **JSON** by default, with the `Content-Type: application/json` header. Every endpoint that returns tabular data also accepts `?format=json|csv|xml|xlsx`: CSV is UTF-8 with a byte-order mark so spreadsheet software decodes currency symbols correctly, XLSX is a genuine Excel workbook with typed dates and full-precision rates, and XML mirrors the JSON field names for ERP and legacy pipelines. An unsupported value returns HTTP 400 before authentication is evaluated.

A typical successful response structure:

```
{
"rate": 0.9234,
"source": "USD",
"target": "EUR",
"time": "2026-04-12T14:30:00Z"
}
```

1.6 Timestamp Format

All timestamps returned in API responses use the **ISO 8601** format:

```
YYYY-MM-DDThh:mm:ssZ
```

The "Z" suffix indicates UTC time. For historical data endpoints, timestamps include millisecond precision where available.

1.7 Package Limits

The number of API requests available to you is determined by the service package you selected at sign-up. Each rate returned counts as one API request toward your monthly package limit.

Package	Monthly Requests	Rate Limit	Price
Free	300	10 req/min	€0/month
Small	5,000	10 req/min	€4.99/month (€49.99/yr)
Medium	10,000	Unlimited	€9.99/month (€99.99/yr)
Large	100,000	Unlimited	€49.99/month (€499/yr)
Custom	Negotiable	Unlimited	Contact us

When you request multiple target currencies in a single query, each currency counts as one request. For example, converting USD to EUR, GBP, and JPY counts as **3 requests**.

Historical data is metered by volume on the same principle: **one API call per currency (series) per month of data**. A one-month single-pair series costs 1 call; a one-year single-pair series costs 12; a whole-table history download covering 30 currencies for a year costs about 360. Each history response reports the charge in its **billed_requests** field.

When you have exhausted your monthly allowance, further requests will return **429 Too Many Requests** until your new monthly period begins.

Yearly billing includes 2 months free. The Large plan includes 3 API keys; all other plans include 1. Historical rates, central bank rate history, and commercial use require a paid plan. The latest central bank rate tables are available on every plan, including Free. See https://allratestoday.com/pricing for current details.

1.7.1 Package Limit Headers

You can monitor your usage via the response headers included with every API response:

Header Name	Description
`X-Monthly-Limit`	The maximum number of requests permitted per month for your package.
`X-Monthly-Used`	The number of requests used in the current monthly period.
`X-Monthly-Resets`	The ISO 8601 timestamp when your monthly usage counter resets.
`X-RateLimit-Limit`	The maximum number of requests permitted per rate-limit window.
`X-RateLimit-Remaining`	The number of requests remaining in the current rate-limit window.
`X-RateLimit-Reset`	The UTC epoch timestamp when the rate-limit window resets.

1.8 Rate Limiting

All requests to the API are subject to rate limiting over a time window to ensure fair usage and prevent excessive requests due to technical errors.

If you exceed the rate limit, you will receive:

```
HTTP/1.1 429 Too Many Requests

{
"error": "Monthly API request limit reached",
"limit": 300,
"used": 300,
"resetsAt": "2026-05-01T00:00:00Z",
"upgrade": "https://allratestoday.com/pricing"
}
```

1.9 CORS Support

The keyless endpoints send `Access-Control-Allow-Origin: *` and can be called from browser JavaScript on any site: `/api/open/central-bank/{bank}`, `/api/open/central-banks`, `/api/open/geo`, `/api/v1/symbols` and the `/api/compat/...` endpoints.

Keyed endpoints (`/api/v1/rates`, `/api/historical-rates` and `/api/v1/central-bank/...`) do not send permissive CORS headers. An API key in browser code is public, so call these from your server and expose your own endpoint to your frontend.

1.10 Error Handling

The API uses standard HTTP status codes to indicate success or failure. All error responses include a JSON body with an **error** field describing the issue.

HTTP Status	Code	Message
200 OK	—	Request succeeded. Response body contains the requested data.
400 Bad Request	400	Missing or invalid query parameters (e.g., invalid currency code).
401 Unauthorized	401	Missing, invalid, or disabled API key.
403 Forbidden	403	Account disabled or suspended, or the endpoint requires a paid plan (e.g., historical market or central bank rates on the Free plan or a trial key; code historical_requires_paid_plan).
429 Too Many Requests	429	Monthly package limit or rate limit exceeded.
500 Internal Server Error	500	Unexpected server error. Please retry or contact support.

2 API Endpoints

This section describes each available API endpoint in detail, including the HTTP method, URL pattern, parameters, example requests, and response formats.

2.1 Get Latest Exchange Rates

Returns the current live exchange rate for one base currency against one or more target currencies. Supports multiple targets in a single request via comma-separated values.

```
GET /api/v1/rates?source={FROM}&amp;target={TO}
```

Example — Single target currency:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/rates?source=USD&amp;target=EUR"
```

2.1.1 Response

```
HTTP/1.1 200 OK
Content-Type: application/json
X-Monthly-Limit: 300
X-Monthly-Used: 42

[
{
"rate": 0.9234,
"source": "USD",
"target": "EUR",
"time": "2026-04-12T14:30:00Z"
}
]
```

Example — Multiple target currencies:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/rates?source=USD&amp;target=EUR,GBP,JPY,CAD"
```

2.1.2 Response (Multiple Targets)

```
[
{ "rate": 0.9234, "source": "USD", "target": "EUR", "time": "2026-04-12T14:30:00Z" },
{ "rate": 0.7891, "source": "USD", "target": "GBP", "time": "2026-04-12T14:30:00Z" },
{ "rate": 151.42, "source": "USD", "target": "JPY", "time": "2026-04-12T14:30:00Z" },
{ "rate": 1.3567, "source": "USD", "target": "CAD", "time": "2026-04-12T14:30:00Z" }
]
```

Note: Requesting 4 target currencies counts as 4 API requests toward your monthly limit.

2.1.3 Parameters

Name	Type	Required	Description
`source`	String	Yes	The base currency ISO 4217 code (e.g., USD, EUR, GBP). If omitted, USD is assumed.

Name	Type	Required	Description
`target`	String	Yes	One or more target currency ISO 4217 codes, comma-separated (e.g., EUR,GBP,JPY).
`time`	String	No	Specific time for the rate query. UTC time in ISO 8601 format.

2.1.4 Response Fields

Name	Type	Required	Description
`rate`	Number	—	The mid-market exchange rate from source to target currency.
`source`	String	—	The base currency ISO 4217 code.
`target`	String	—	The target currency ISO 4217 code.
`time`	String	—	ISO 8601 timestamp of when the rate was recorded.

2.2 Get Single Pair Rate

A simplified, unauthenticated endpoint that returns the current exchange rate for a single currency pair. Designed for quick lookups and lightweight integrations.

```
GET /api/rate?source={FROM}&amp;target={TO}
```

Example:

```
curl "https://allratestoday.com/api/rate?source=EUR&amp;target=LKR"
```

2.2.1 Response

```
HTTP/1.1 200 OK
Content-Type: application/json

{
"rate": 415.28,
"source": "wise"
}
```

2.2.2 Parameters

Name	Type	Required	Description
`source`	String	Yes	The base currency ISO 4217 code (e.g., USD, EUR, GBP).
`target`	String	Yes	The target currency ISO 4217 code (e.g., LKR, INR, JPY).

Note: This endpoint does not require authentication but is intended for simple, single-pair lookups only. For production use with multiple currencies, use the authenticated /api/v1/rates endpoint.

2.3 Get Historical Rates

Returns historical exchange rate data for a currency pair over a specified time period. Paid plans only (Free plan and trial keys receive 403). Supports preset periods of 1 day, 7 days, 30 days, and 1 year. Billed by volume: periods of one month or less count as 1 API call; the 1-year period is billed as 12 calls (one per month of data).

```
GET /api/historical-rates?source={FROM}&amp;target={TO}&amp;period={PERIOD}
```

Example — 30-day history:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/historical-rates?source=EUR&amp;target=LKR&amp;period=30d"
```

2.3.1 Response

```
HTTP/1.1 200 OK
Content-Type: application/json

{
"source": "EUR",
"target": "LKR",
"data": [
{
"date": "2026-03-13T00:00:00.000Z",
```

```
"rate": 358.4521,
"timestamp": 1710288000000
},
{
"date": "2026-03-14T00:00:00.000Z",
"rate": 360.1214,
"timestamp": 1710374400000
},
...
],
"source_api": "wise",
"period": "30d"
}
```


2.3.2 Parameters

Name	Type	Required	Description
`source`	String	Yes	The base currency ISO 4217 code.
`target`	String	Yes	The target currency ISO 4217 code.
`period`	String	No	The time period for historical data. Accepted values: **1d** (1 day — 24 hourly data points), **7d** (7 days — daily, default), **30d** (30 days — daily), **1y** (1 year — sampled every 7 days).

2.3.3 Response Fields

Name	Type	Required	Description
`source`	String	—	The base currency ISO 4217 code.
`target`	String	—	The target currency ISO 4217 code.
`data`	Array	—	Array of historical data points, each containing date, rate, and timestamp.
`data[].date`	String	—	ISO 8601 date string for the data point.
`data[].rate`	Number	—	The mid-market exchange rate at this date.
`data[].timestamp`	Number	—	Unix timestamp in milliseconds for the data point.
`source_api`	String	—	The upstream data provider identifier (e.g., "wise").
`period`	String	—	The period requested (1d, 7d, 30d, or 1y).

2.4 Get Historical Rate Period

Returns daily exchange rates for one currency pair within a custom date range, newest first. Use the **from** and **to** date parameters (the **to** date is exclusive) with **group=day**. Paid plans only. A range covers at most 366 days per request (longer ranges return 400) and is billed one API call per 31 days of data, so a full year costs 12.

```
GET
/api/v1/rates?source={FROM}&amp;target={TO}&amp;from={START_DATE}&amp;to={END_DATE}
```

Example — Custom date range:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/rates?source=USD&amp;target=CAD&amp;from=2026-01-01&
amp;to=2026-04-01&amp;group=day"
```

2.4.1 Response

```
HTTP/1.1 200 OK
Content-Type: application/json

[
{
"rate": 1.3421,
"source": "USD",
"target": "CAD",
"time": "2026-01-01T00:00:00Z"
},
{
"rate": 1.3456,
"source": "USD",
"target": "CAD",
"time": "2026-01-02T00:00:00Z"
},
...
]
```

2.4.2 Parameters

Name	Type	Required	Description
`source`	String	Yes	The base currency ISO 4217 code.
`target`	String	Yes	One target currency ISO 4217 code (several targets return one row each, not a series).
`from`	String	Yes	Start date in YYYY-MM-DD format (ISO 8601).
`to`	String	Yes	End date in YYYY-MM-DD format (ISO 8601).
`group`	String	No	Grouping interval for the data points (e.g., day, hour).

2.5 Get Available Currencies

Returns a complete list of all supported currency codes, names, and symbols. This endpoint is cached for 24 hours and does not count toward your monthly API limit.

```
GET /api/v1/symbols
```

Example:

```
curl "https://allratestoday.com/api/v1/symbols"
```

2.5.1 Response

```
HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: public, max-age=86400

{
"currencies": [
{ "code": "AED", "name": "UAE Dirham", "symbol": "AED" },
{ "code": "AUD", "name": "Australian Dollar", "symbol": "A$" },
{ "code": "BDT", "name": "Bangladeshi Taka", "symbol": "Tk" },
{ "code": "BRL", "name": "Brazilian Real", "symbol": "R$" },
{ "code": "CAD", "name": "Canadian Dollar", "symbol": "CA$" },
{ "code": "EUR", "name": "Euro", "symbol": "€" },
{ "code": "GBP", "name": "British Pound", "symbol": "£" },
{ "code": "INR", "name": "Indian Rupee", "symbol": "■" },
{ "code": "JPY", "name": "Japanese Yen", "symbol": "¥" },
{ "code": "USD", "name": "US Dollar", "symbol": "$" },
...
],
"count": 160
}
```

Note: This endpoint does not require authentication and is cached for 24 hours.

2.5.2 Response Fields

Name	Type	Required	Description
`currencies`	Array	—	Array of currency objects.
`currencies[].code`	String	—	ISO 4217 three-letter currency code.
`currencies[].name`	String	—	Full English name of the currency.
`currencies[].symbol`	String	—	The currency symbol (e.g., $, €, £, ¥).
`count`	Number	—	Total number of currencies returned.

3 Central Bank Rates API

AllRatesToday publishes the **official reference exchange rates** of 121 central banks, exactly as each bank publishes them — including the US Federal Reserve (H.10), European Central Bank, Bank of England, People's Bank of China (CFETS), Bank of Japan, and 26 more. The dataset covers 11M+ historical data points with daily updates following each bank's publication schedule.

All central bank endpoints require Bearer token authentication. The **latest** rate table and the **availability** calendar are included on every plan (including Free); **historical dates** and **time series** require a paid plan. Endpoints that return tabular data support `format=csv`, `format=xlsx` and `format=xml` in addition to the default JSON.

For the complete list of covered banks and their bank codes, see Appendix B or call `GET /api/v1/central-banks`.

3.1 List Covered Central Banks

Returns metadata for every covered central bank: bank code, name, country, home currency, latest available rate date, and endpoint URLs.

```
GET /api/v1/central-banks
```

Example:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/central-banks"
```

3.2 Get Latest Official Central Bank Rates

Returns the most recent full rate table as published by the central bank. Pass **source** and **target** together for a single-pair lookup — when the pair is not directly published by the bank, the rate is cross-computed via the bank's home currency and flagged with `derived: true`. Available on all plans.

```
GET /api/v1/central-bank/{bank}/latest
```

Example — Full ECB table:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/central-bank/ecb/latest"
```

Example — Single pair from the Central Bank of Sri Lanka:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/central-bank/cbsl/latest?source=USD&amp;target=LKR"
```

3.2.1 Parameters

Name	Type	Required	Description
`bank`	String	Yes	Bank code from /api/v1/central-banks (e.g., fed, ecb, boe, cbsl). Path parameter.
`source`	String	No	Pair mode: source currency ISO 4217 code (requires target).

Name	Type	Required	Description
target	String	No	Pair mode: target currency ISO 4217 code (requires source).
format	String	No	Response format: **json** (default), **csv**, **xml** or **xlsx**.

Note: Responses include a rate_date field — the date the bank actually published the table. Unknown bank codes return 404.

3.3 Get Official Rates for a Date

Returns the bank's published rate table for a specific date. For weekends and holidays, the API returns the most recent published table on or before the requested date (indicated by the **rate_date** field). Paid plans only.

```
GET /api/v1/central-bank/{bank}/{date}
```

Example:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/central-bank/ecb/2026-06-30?source=USD&amp;target=EUR"
```

3.3.1 Parameters

Name	Type	Required	Description
bank	String	Yes	Bank code from /api/v1/central-banks. Path parameter.
date	String	Yes	Date in YYYY-MM-DD format. Path parameter.
source	String	No	Pair mode: source currency (requires target).
target	String	No	Pair mode: target currency (requires source).

3.4 Get Central Bank Rate Time Series

Returns a daily series of official rates over a date range (defaults to the last year, maximum 5,000 rows per request). Use **symbol** for all pairs involving one currency, or **source** + **target** for a single-pair series. Paid plans only. Billed by volume: one API call per month of data covered (a one-year series is 12 calls); the response reports the charge in **billed_requests**.

```
GET /api/v1/central-bank/{bank}/history
```

Example — USD/CAD series from the Bank of Canada:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/central-bank/boc/history?source=USD&amp;target=CAD&amp;from=2026-01-01&amp;to=2026-06-30"
```

3.4.1 Parameters

Name	Type	Required	Description
bank	String	Yes	Bank code from /api/v1/central-banks. Path parameter.

Name	Type	Required	Description
`symbol`	String	No	All pairs involving this currency. Required unless source+target given.
`source`	String	No	Pair mode: source currency (requires target).
`target`	String	No	Pair mode: target currency (requires source).
`from`	String	No	Start date in YYYY-MM-DD format. Defaults to one year ago.
`to`	String	No	End date in YYYY-MM-DD format. Defaults to today.
`format`	String	No	Response format: **json** (default), **csv**, **xml** or **xlsx**.

3.5 Get Publication Calendar (Availability)

Returns the dates on which the bank actually published a rate table, for a calendar year or custom date range (defaults to the last year). Weekends, holidays, and weekly-cadence gaps (e.g., the Federal Reserve H.10 is weekly) appear as missing dates. Available on all plans.

```
GET /api/v1/central-bank/{bank}/availability
```

Example:

```
curl -H "Authorization: Bearer art_live_your_api_key" \
"https://allratestoday.com/api/v1/central-bank/cbsl/availability?year=2026"
```

3.5.1 Parameters

Name	Type	Required	Description
`bank`	String	Yes	Bank code from /api/v1/central-banks. Path parameter.
`year`	Integer	No	Whole calendar year (overrides from/to).
`from`	String	No	Start date in YYYY-MM-DD format.
`to`	String	No	End date in YYYY-MM-DD format.
`format`	String	No	Response format: **json** (default), **csv**, **xml** or **xlsx**.

4 SDKs & Libraries

AllRatesToday provides official SDKs in multiple programming languages. These SDKs handle authentication, request construction, error handling, and response parsing, allowing you to integrate exchange rate data with minimal code.

4.1 JavaScript / TypeScript SDK

Package: `@allratestoday/sdk`
Registry: npm — https://www.npmjs.com/package/@allratestoday/sdk
Requirements: Node.js 18+ or modern browser (ES modules)
Dependencies: Zero runtime dependencies

Installation

```
npm install @allratestoday/sdk
```

Quick Start

```
import { AllRatesToday } from '@allratestoday/sdk';

const client = new AllRatesToday('art_live_your_api_key');

// Get latest rate
const rates = await client.latest({ base: 'USD', symbols: ['EUR', 'GBP'] });
console.log(rates.EUR); // 0.9234

// Convert currency
const result = await client.convert({
from: 'USD',
to: 'EUR',
amount: 1000
});
console.log(result.converted); // 923.40

// Get historical rates
const history = await client.historical({
base: 'USD',
date: '2025-01-15'
});
console.log(history.EUR); // 0.9180
```

4.2 Python SDK

Package: `allratestoday`
Registry: PyPI — https://pypi.org/project/allratestoday/
Requirements: Python 3.7+
Dependencies: Zero — uses only stdlib (urllib + json)

Installation

```
pip install allratestoday
```

Quick Start

```
from allratestoday import AllRatesToday

client = AllRatesToday('art_live_your_api_key')

# Get latest rates
rates = client.latest(base='USD')
print(rates['EUR']) # 0.9234

# Get historical rates
history = client.historical(base='USD', date='2025-01-15')
print(history['EUR']) # 0.9180

# List available currencies
currencies = client.currencies()
print(len(currencies)) # 160+
```

4.3 PHP SDK

Package: `allratestoday/sdk`
Registry: Packagist — https://packagist.org/packages/allratestoday/sdk
Requirements: PHP 7.4+, cURL extension, JSON extension

Installation

```
composer require allratestoday/sdk
```

Quick Start

```
&lt;?php
use AllRatesToday\Client;

$client = new Client('art_live_your_api_key');

// Get latest rates
$rates = $client->latest(['base' => 'USD']);
echo $rates->EUR; // 0.9234

// Convert currency
$result = $client->convert([
'from' => 'USD',
'to' => 'EUR',
'amount' => 1000
]);
echo $result->converted; // 923.40
```

4.4 React SDK

Package: `react-currency-localizer-realtime`
Registry: npm — https://www.npmjs.com/package/react-currency-localizer-realtime
Requirements: React 16.8+ (hooks support)
Bundle Size: ~10KB minified (~4KB gzipped) — zero runtime dependencies beyond React

Installation

```
npm install react-currency-localizer-realtime
```

Quick Start

```
import { useCurrencyLocalizer } from
'react-currency-localizer-realtime';

function PricingCard({ priceUSD }) {
const { convert, userCurrency, loading }
= useCurrencyLocalizer({
apiKey: 'art_live_your_api_key',
baseCurrency: 'USD'
});

if (loading) return &lt;Spinner /&gt;;

return (
&lt;div className="price"&gt;
{convert(priceUSD)} {userCurrency}
&lt;/div&gt;
);
}
```

Key features: Automatic IP-based currency detection, built-in rate caching, SSR compatible (Next.js, Remix), full TypeScript support.

4.5 Rust SDK

Package: `allratestoday`
Registry: crates.io — https://crates.io/crates/allratestoday
Documentation: https://docs.rs/allratestoday

Installation

```
cargo add allratestoday
```

Quick Start

```
use allratestoday::AllRatesToday;

fn main() {
let client = AllRatesToday::new("art_live_your_api_key");

// Get latest rates
let rates = client.latest("USD", Some(&amp;["EUR", "GBP"])).unwrap();
println!("{:?}", rates);

// Convert currency
let result = client.convert("USD", "EUR", 100.0).unwrap();
println!("100 USD = {:?} EUR", result.result);
}
```

4.6 MCP Server (AI Assistants)

Package: `@allratestoday/mcp-server`
Registry: npm — https://www.npmjs.com/package/@allratestoday/mcp-server
Protocol: Model Context Protocol (MCP) 1.x, stdio transport

A Model Context Protocol server that gives AI assistants — Claude Code, Claude Desktop, Cursor, Windsurf, and any MCP-compatible client — direct access to live exchange rates, historical data, and multi-currency lookups from the AllRatesToday API.

Setup (Claude Code example)

```
claude mcp add allratestoday \
--env ALLRATESTODAY_API_KEY=art_live_your_api_key \
-- npx -y @allratestoday/mcp-server
```

4.7 Per-Bank npm SDKs

Each of the 93 covered central banks has a dedicated, zero-dependency npm package that wraps the Central Bank Rates API for that bank — for example `ecb-exchange-rate`, `fed-exchange-rate`, `boe-exchange-rate`, and `boj-exchange-rate`. See Appendix B for the full package list.

Installation & Quick Start (ECB example)

```
npm install ecb-exchange-rate

import { getRate, getLatestRates } from 'ecb-exchange-rate';

// One pair at the official ECB rate
const pair = await getRate('USD', 'EUR', { apiKey: 'art_live_...' });
console.log(pair.rate, pair.rate_date);

// The bank's full published table
const table = await getLatestRates({ apiKey: 'art_live_...' });
console.log(table.rate_date, table.rates.length);
```

5 Appendix A — Supported Currencies

AllRatesToday supports over 160 world currencies. Below is a representative list of the most commonly used currencies available through the API. For a complete, up-to-date list, use the `GET /api/v1/symbols` endpoint.

Code	Currency Name	Code	Currency Name	Code	Currency Name
AED	UAE Dirham	ARS	Argentine Peso	AUD	Australian Dollar
BDT	Bangladeshi Taka	BGN	Bulgarian Lev	BRL	Brazilian Real
CAD	Canadian Dollar	CHF	Swiss Franc	CNY	Chinese Yuan
COP	Colombian Peso	CZK	Czech Koruna	DKK	Danish Krone
EGP	Egyptian Pound	EUR	Euro	GBP	British Pound Sterling
GHS	Ghanaian Cedi	HKD	Hong Kong Dollar	HUF	Hungarian Forint
IDR	Indonesian Rupiah	ILS	Israeli Shekel	INR	Indian Rupee
JPY	Japanese Yen	KES	Kenyan Shilling	KRW	South Korean Won
LKR	Sri Lankan Rupee	MAD	Moroccan Dirham	MXN	Mexican Peso
MYR	Malaysian Ringgit	NGN	Nigerian Naira	NOK	Norwegian Krone
NZD	New Zealand Dollar	PEN	Peruvian Sol	PHP	Philippine Peso
PKR	Pakistani Rupee	PLN	Polish Zloty	RON	Romanian Leu
RUB	Russian Ruble	SAR	Saudi Riyal	SEK	Swedish Krona
SGD	Singapore Dollar	THB	Thai Baht	TRY	Turkish Lira
TWD	Taiwan Dollar	TZS	Tanzanian Shilling	UAH	Ukrainian Hryvnia
UGX	Ugandan Shilling	USD	US Dollar	VND	Vietnamese Dong
ZAR	South African Rand	ZMW	Zambian Kwacha		

This is a partial list. The API supports 160+ currencies in total. Use the /api/v1/symbols endpoint to retrieve the complete, current list of supported currencies.

6 Appendix B — Covered Central Banks

The Central Bank Rates API (Section 3) covers the following 121 central banks. Use the **Code** column as the {bank} path parameter. Each bank also has a dedicated npm SDK (Section 4.7).

Code	Central Bank	Country	Home Ccy	npm Package
`fed`	US Federal Reserve (H.10)	United States	USD	`fed-exchange-rate`
`boe`	Bank of England	United Kingdom	GBP	`boe-exchange-rate`
`pboc`	People's Bank of China (CFETS)	China	CNY	`pboc-exchange-rate`
`rba`	Reserve Bank of Australia	Australia	AUD	`rba-exchange-rate`
`cbr`	Bank of Russia	Russia	RUB	`cbr-exchange-rate`
`tcmb`	Central Bank of Türkiye	Türkiye	TRY	`tcmb-exchange-rate`
`bcb`	Banco Central do Brasil (PTAX)	Brazil	BRL	`bcb-exchange-rate`
`bi`	Bank Indonesia	Indonesia	IDR	`bank-indonesia-exchange-rate`
`rbi`	Reserve Bank of India	India	INR	`rbi-exchange-rate`
`boc`	Bank of Canada	Canada	CAD	`boc-exchange-rate`
`bnm`	Bank Negara Malaysia	Malaysia	MYR	`bnm-exchange-rate`
`cbsl`	Central Bank of Sri Lanka	Sri Lanka	LKR	`cbsl-exchange-rate`
`cnb`	Czech National Bank	Czech Republic	CZK	`czech-national-bank-exchange-rate`
`ecb`	European Central Bank	Eurozone	EUR	`ecb-exchange-rate`
`nbp`	Narodowy Bank Polski	Poland	PLN	`nbp-exchange-rate`
`norges`	Norges Bank	Norway	NOK	`norges-bank-exchange-rate`
`riksbank`	Sveriges Riksbank	Sweden	SEK	`riksbank-exchange-rate`
`cbn`	Central Bank of Nigeria	Nigeria	NGN	`cbn-exchange-rate`
`cbk`	Central Bank of Kuwait	Kuwait	KWD	`cbk-exchange-rate`
`cbb`	Central Bank of Bahrain	Bahrain	BHD	`cbb-exchange-rate`
`cbar`	Central Bank of Azerbaijan	Azerbaijan	AZN	`cbar-exchange-rate`
`cbbh`	Central Bank of Bosnia and Herzegovina	Bosnia and Herzegovina	BAM	`cbbh-exchange-rate`
`bnb`	Bulgarian National Bank	Bulgaria	EUR	`bnb-exchange-rate`
`bcrp`	Central Reserve Bank of Peru	Peru	PEN	`bcrp-exchange-rate`
`boj`	Bank of Japan	Japan	JPY	`boj-exchange-rate`
`hkma`	Hong Kong Monetary Authority	Hong Kong	HKD	`hkma-exchange-rate`
`boi`	Bank of Israel	Israel	ILS	`bank-of-israel-exchange-rate`

Code	Central Bank	Country	Home Ccy	npm Package
bcra	Central Bank of Argentina	Argentina	ARS	bcra-exchange-rate
sarb	South African Reserve Bank	South Africa	ZAR	sarb-exchange-rate
snb	Swiss National Bank	Switzerland	CHF	snb-exchange-rate
rbnz	Reserve Bank of New Zealand	New Zealand	NZD	rbnz-exchange-rate
bok	Bank of Korea	South Korea	KRW	bok-exchange-rate
bnr	National Bank of Romania	Romania	RON	bnr-exchange-rate
nbu	National Bank of Ukraine	Ukraine	UAH	nbu-exchange-rate
banxico	Banco de México	Mexico	MXN	banxico-exchange-rate
bsp	Bangko Sentral ng Pilipinas	Philippines	PHP	bsp-exchange-rate
bcrd	Banco Central de la República Dominicana	Dominican Republic	DOP	bcrd-exchange-rate
bcp	Banco Central del Paraguay	Paraguay	PYG	bcp-exchange-rate
nbc	National Bank of Cambodia	Cambodia	KHR	nbc-cambodia-exchange-rate
bog	Bank of Ghana	Ghana	GHS	bank-of-ghana-exchange-rate
cbe	Central Bank of Egypt	Egypt	EGP	cbe-exchange-rate
boa	Bank of Albania	Albania	ALL	bank-of-albania-exchange-rate
banguat	Banco de Guatemala	Guatemala	GTQ	banguat-exchange-rate
bom	Bank of Mongolia	Mongolia	MNT	bank-of-mongolia-exchange-rate
mas	Monetary Authority of Singapore	Singapore	SGD	mas-exchange-rate
bdi	Banca d'Italia	Italy	EUR	bdi-exchange-rate
dnb	Danmarks Nationalbank	Denmark	DKK	danmarks-nationalbank-exchange-rate
sfc	Superintendencia Financiera de Colombia (TRM)	Colombia	COP	colombia-trm-exchange-rate
nbk	National Bank of Kazakhstan	Kazakhstan	KZT	nbk-exchange-rate
nbg	National Bank of Georgia	Georgia	GEL	nbg-exchange-rate
cbu	Central Bank of Uzbekistan	Uzbekistan	UZS	cbu-exchange-rate
mnb	Magyar Nemzeti Bank	Hungary	HUF	magyar-nemzeti-bank-exchange-rate
cbi	Central Bank of Iceland	Iceland	ISK	central-bank-of-iceland-exchange-rate
cbuae	Central Bank of the UAE	United Arab Emirates	AED	cbuae-exchange-rate
sbp	State Bank of Pakistan	Pakistan	PKR	sbp-exchange-rate

Code	Central Bank	Country	Home Ccy	npm Package
nrb	Nepal Rastra Bank	Nepal	NPR	nrb-exchange-rate
cbj	Central Bank of Jordan	Jordan	JOD	cbj-exchange-rate
bb	Bangladesh Bank	Bangladesh	BDT	bangladesh-bank-exchange-rate
cbo	Central Bank of Oman	Oman	OMR	cbo-exchange-rate
pma	Palestine Monetary Authority	Palestine	ILS	palestine-monetary-authority-exchange-rate
cbm	Central Bank of Myanmar	Myanmar	MMK	cbm-exchange-rate
cbc	Central Bank of the Republic of China (Taiwan)	Taiwan	TWD	taiwan-central-bank-exchange-rate
bot	Bank of Thailand	Thailand	THB	bank-of-thailand-exchange-rate
amcm	Monetary Authority of Macao	Macao	MOP	amcm-exchange-rate
sbv	State Bank of Vietnam	Vietnam	VND	sbv-exchange-rate
bol	Bank of the Lao PDR	Laos	LAK	bank-of-laos-exchange-rate
bcv	Central Bank of Venezuela	Venezuela	VES	central-bank-of-venezuela-exchange-rate
bcch	Central Bank of Chile	Chile	CLP	central-bank-of-chile-exchange-rate
bcu	Central Bank of Uruguay	Uruguay	UYU	bcu-exchange-rate
bcbol	Banco Central de Bolivia	Bolivia	BOB	banco-central-de-bolivia-exchange-rate
cbvs	Central Bank of Suriname	Suriname	SRD	central-bank-of-suriname-exchange-rate
bogy	Bank of Guyana	Guyana	GYD	bank-of-guyana-exchange-rate
sama	Saudi Central Bank (SAMA)	Saudi Arabia	SAR	sama-exchange-rate
bam	Bank Al-Maghrib	Morocco	MAD	bank-al-maghrib-exchange-rate
nbs	National Bank of Serbia	Serbia	RSD	national-bank-of-serbia-exchange-rate
nbrb	National Bank of the Republic of Belarus	Belarus	BYN	nbrb-exchange-rate
qcb	Qatar Central Bank	Qatar	QAR	qcb-exchange-rate
cbiq	Central Bank of Iraq	Iraq	IQD	central-bank-of-iraq-exchange-rate
bda	Bank of Algeria	Algeria	DZD	bank-of-algeria-exchange-rate
bccr	Banco Central de Costa Rica	Costa Rica	CRC	bccr-exchange-rate

Code	Central Bank	Country	Home Ccy	npm Package
bceao	Central Bank of West African States (BCEAO)	West Africa (WAEMU)	XOF	bceao-exchange-rate
nbe	National Bank of Ethiopia	Ethiopia	ETB	national-bank-of-ethiopia-exchange-rate
cbke	Central Bank of Kenya	Kenya	KES	central-bank-of-kenya-exchange-rate
nbm	National Bank of Moldova	Moldova	MDL	nbm-exchange-rate
nbrm	National Bank of North Macedonia	North Macedonia	MKD	nbrm-exchange-rate
cba	Central Bank of Armenia	Armenia	AMD	cba-exchange-rate
beac	Bank of Central African States (BEAC)	Central Africa (CEMAC)	XAF	beac-exchange-rate
bct	Central Bank of Tunisia	Tunisia	TND	bct-exchange-rate
bnrw	National Bank of Rwanda	Rwanda	RWF	national-bank-of-rwanda-exchange-rate
bna	National Bank of Angola	Angola	AOA	bna-exchange-rate
botz	Bank of Tanzania	Tanzania	TZS	bank-of-tanzania-exchange-rate
bou	Bank of Uganda	Uganda	UGX	bou-exchange-rate
bojm	Bank of Jamaica	Jamaica	JMD	bank-of-jamaica-exchange-rate
bcn	Banco Central de Nicaragua	Nicaragua	NIO	bcn-exchange-rate
bch	Banco Central de Honduras	Honduras	HNL	bch-exchange-rate
cbtt	Central Bank of Trinidad and Tobago	Trinidad and Tobago	TTD	cbtt-exchange-rate
bmu	Bank of Mauritius	Mauritius	MUR	bmu-exchange-rate
bdm	Banco de Moçambique	Mozambique	MZN	bdm-exchange-rate
bob	Bank of Botswana	Botswana	BWP	bob-exchange-rate
boz	Bank of Zambia	Zambia	ZMW	boz-exchange-rate
rbm	Reserve Bank of Malawi	Malawi	MWK	rbm-exchange-rate
nbkr	National Bank of the Kyrgyz Republic	Kyrgyzstan	KGS	nbkr-exchange-rate
lb	Bank of Lithuania	Lithuania	EUR	bank-of-lithuania-exchange-rate
bcc	Banque Centrale du Congo	DR Congo	CDF	banque-centrale-du-congo-exchange-rate
cbbd	Central Bank of Barbados	Barbados	BBD	central-bank-of-barbados-exchange-rate

Code	Central Bank	Country	Home Ccy	npm Package
cbl	Central Bank of Libya	Libya	LYD	central-bank-of-libya-exchange-rate
cbg	Central Bank of The Gambia	Gambia	GMD	central-bank-of-the-gambia-exchange-rate
brh	Banque de la République d'Haïti	Haiti	HTG	brh-exchange-rate
rma	Royal Monetary Authority of Bhutan	Bhutan	BTN	rma-bhutan-exchange-rate
nbt	National Bank of Tajikistan	Tajikistan	TJS	national-bank-of-tajikistan-exchange-rate
cbbz	Central Bank of Belize	Belize	BZD	central-bank-of-belize-exchange-rate
rbf	Reserve Bank of Fiji	Fiji	FJD	reserve-bank-of-fiji-exchange-rate
mma	Maldives Monetary Authority	Maldives	MVR	maldives-monetary-authority-exchange-rate
bcm	Banque Centrale de Mauritanie	Mauritania	MRU	banque-centrale-de-mauritanie-exchange-rate
rbz	Reserve Bank of Zimbabwe	Zimbabwe	ZWG	reserve-bank-of-zimbabwe-exchange-rate
bon	Bank of Namibia	Namibia	NAD	bank-of-namibia-exchange-rate
bfm	Central Bank of Madagascar (BFM)	Madagascar	MGA	central-bank-of-madagascar-exchange-rate

Coverage grows over time. Use the /api/v1/central-banks endpoint to retrieve the current list of covered banks, their latest available dates, and endpoint URLs.

7 Appendix C — Error Codes Reference

The following table lists all possible error responses from the API:

HTTP Status	Code	Message
400	INVALID_PARAMS	Missing or invalid query parameters. Check that source and target are valid ISO 4217 currency codes.
400	INVALID_CURRENCY	The specified currency code is not supported. Use /api/v1/symbols for valid codes.
400	INVALID_DATE	The date parameter is not in valid YYYY-MM-DD format or is out of range.
400	INVALID_PERIOD	The period parameter must be one of: 1d, 7d, 30d, 1y.
401	AUTH_REQUIRED	No API key provided. Include an Authorization: Bearer header.
401	INVALID_KEY	The API key is invalid or has been revoked.
401	KEY_DISABLED	The API key has been disabled by an administrator.
401	ACCOUNT_DISABLED	Your account has been disabled. Contact support.
401	EMAIL_NOT_VERIFIED	Your email address has not been verified. Check your inbox.
403	PAID_FEATURE	This endpoint requires a paid plan (e.g., historical market rates, central bank historical dates and time series on the Free plan).
404	UNKNOWN_BANK	The central bank code is not covered. Use /api/v1/central-banks for valid codes.

HTTP Status	Code	Message
429	MONTHLY_LIMIT	Monthly API request limit exceeded. Upgrade your plan or wait for reset.
429	RATE_LIMIT	Too many requests in the current time window. Slow down.
500	INTERNAL_ERROR	An unexpected server error occurred. Please retry.
500	UPSTREAM_ERROR	The upstream data provider returned an error. Please retry.
503	SERVICE_UNAVAILABLE	The service is temporarily unavailable for maintenance.

Error Response Format:

```
{
"error": "Human-readable error description",
"code": "MACHINE_READABLE_CODE",
"docs": "https://allratestoday.com/docs"
}
```

8 Appendix D — Package Limit Alerts

When you are nearing your monthly package limit, AllRatesToday will send email alerts to the email address associated with your account. Notifications are sent at the following thresholds:

Threshold	Alert Level	Description
75%	Warning	You have used 75% of your monthly API request allowance. Consider monitoring usage or upg
90%	Critical	You have used 90% of your monthly allowance. You are approaching your limit. Upgrade recon
100%	Limit Reached	You have reached your monthly limit. Further API requests will return 429 until the next billing p

You can also monitor your current usage in real-time by checking the **X-Monthly-Used** and **X-Monthly-Limit** response headers returned with every API response, or by visiting your profile dashboard at https://allratestoday.com/profile.

To avoid service interruption, we recommend setting up monitoring on the response headers and upgrading your plan proactively when approaching your limit.

9 Appendix E — Changelog

Version 2.9 — October 2026

• Historical market rates (`time`, `from`/`to` on /api/v1/rates and /api/historical-rates) are now paid-only; Free plan and trial keys receive 403 `historical_requires_paid_plan`.

• Date ranges on /api/v1/rates are billed per 31 days of data (a year costs 12) and capped at 366 days per request.

• The `source` / `source_api` value in rate responses is now `mid-market` (previously `interbank`).

• CORS section corrected: only keyless endpoints are callable from third-party browser pages.

Version 2.8 — September 2026

• Coverage expanded to **121 central banks**: Central Bank of Turkmenistan (about 45 currencies plus the SDR in manat, Monday–Saturday, history since January 2009), Banco Central de Cuba (13 currencies in three official segments — rate types `official`, `public` and `special` — via the bank's own API, history since 19 December 2025), Bank of Sierra Leone (daily reference mid rate for the new leone, 25 currencies) and Banco Central de São Tomé e Príncipe (daily buy/sell/mid bulletin, 14 currencies plus the SDR); the last two build their archives forward from September 2026.

Version 2.7 — September 2026

• Coverage expanded to **121 central banks**: Central Bank of Madagascar (BFM) — the daily closing rate of the interbank foreign-exchange market (MID) for 19 currencies in ariary per unit, history since 22 September 2023.

Version 2.6 — September 2026

• Coverage expanded to **121 central banks**: Banque Centrale de Mauritanie (19-currency daily reference rate, history since December 2023), Reserve Bank of Zimbabwe (daily interbank buy/sell/mid for the ZiG, 34 currencies, since the currency's launch on 8 April 2024) and the Bank of Namibia (daily USD/EUR/GBP/AOA indicative rates; archive builds forward from September 2026).

Version 2.5 — September 2026

• Coverage expanded to **113 central banks**: National Bank of Tajikistan (36 currencies, history since 2010), Central Bank of Belize (buy/sell since 2010), Reserve Bank of Fiji (daily mid-rates since 2001) and the Maldives Monetary Authority (USD reference rate since 2011).

Version 2.4 — September 2026

• Coverage expanded to **109 central banks**: Lithuania (LT list), DR Congo, Barbados, Libya, The Gambia, Haiti and Bhutan, plus the Central American, Caribbean, Indian Ocean and Southern African sources added since 2.3.

Version 2.3 — August 2026

• Coverage expanded to **93 central banks**: Moldova, North Macedonia, Armenia, BEAC (6 CEMAC countries), Tunisia, Rwanda, Angola, Tanzania, Uganda, and Jamaica.

Version 2.2 — August 2026

• Central bank coverage expanded to **83 central banks** (from 31), each with its own npm SDK — Appendix B lists the full catalogue.

• Historical dataset grown to 11M+ central bank rate data points.

Version 2.1 — August 2026

• Volume-based billing for historical data: one API call per currency (series) per month of data. One-month requests still cost 1 call; a one-year single-pair series is 12; whole-table history downloads are billed per currency per month.

• History responses now include a **billed_requests** field reporting the quota charge.

• Additional response formats on every table endpoint: format=json, csv, xml, xlsx.

Version 2.0 — August 2026

• New **Central Bank Rates API**: official reference rates from 31 central banks, including the US Federal Reserve (H.10), ECB, Bank of England, People's Bank of China, and Bank of Japan.

• 5 new endpoints: List Central Banks, Latest Rates, Rates for a Date, Rate Time Series, and Publication Calendar (Availability).

• 2.9M+ historical central bank rate data points; daily updates following each bank's publication schedule.

• Pair mode with cross-rate computation via the bank's home currency, flagged with derived: true.

• CSV export (format=csv) on central bank endpoints.

- New official Rust SDK (crates.io: allratestoday).
- New MCP server for AI assistants (npm: @allratestoday/mcp-server) — Claude Code, Claude Desktop, Cursor, Windsurf.
- 31 dedicated per-bank npm SDKs (e.g., ecb-exchange-rate, fed-exchange-rate, boe-exchange-rate).
- Pricing now in EUR with yearly billing (2 months free): Small €4.99/mo, Medium €9.99/mo, Large €49.99/mo.
- Latest central bank rates available on all plans, including Free; historical central bank data on paid plans.

Version 1.0 — April 2026

- Initial release of the AllRatesToday Currency Data API.
- 5 API endpoints: Latest Rates, Single Pair, Historical Rates, Historical Period, Available Currencies.
- Official SDKs for JavaScript/TypeScript, Python, PHP, and React.
- Support for 160+ world currencies with mid-market rates.
- Mid-market rates aggregated from institutional foreign exchange market data.
- Deployed on Cloudflare Workers (300+ global edge locations).
- Bearer token authentication with API key format: art_live_*.
- Monthly package limits with email alert notifications at 75%, 90%, and 100%.
- Rate limiting with X-RateLimit-* response headers.
- CORS support for browser-based applications.

AllRatesToday

Website: https://allratestoday.com

Register: https://allratestoday.com/register

Documentation: https://allratestoday.com/developers

Pricing: https://allratestoday.com/pricing

API Status: https://allratestoday.com/status

GitHub: https://github.com/AllRates-Today

NPM: https://www.npmjs.com/package/@allratestoday/sdk

PyPI: https://pypi.org/project/allratestoday/

Packagist: https://packagist.org/packages/allratestoday/sdk

Crates.io: https://crates.io/crates/allratestoday

MCP Server: https://www.npmjs.com/package/@allratestoday/mcp-server

Central Bank Rates: https://allratestoday.com/central-bank-rates-api/

Contact: https://allratestoday.com/contact

LinkedIn: https://www.linkedin.com/company/allratestoday/

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AllRatesToday Currency Data API Specifications — Version 2.8 — September 2026